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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Field Sample Permit: The study was carried out under license (permit no. 311.4-si and 311.5-gr, Landratsamt Starnberg, Germany).
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
Using a dummy female, we collected natural ejaculates from eight brothers from two families (mtDNA haplotype A, Nejaculates = 9, from males A1 to A3 mtDNA haplotype B, Nejaculates = 11, from males B2 to B6).
Subject Demographics
Age: Males were 100 to 1,036 d old.
Randomization
All phylogenetic trees were built using RAxML-NG v1.0.2 (46) assuming a general time-reversible model and a discrete gamma model of rate heterogeneity with 100 randomized parsimony starting trees and 1,000 bootstrap replicates (for details, see Data Accessibility).
Blinding
not detected.
Power Analysis
not detected.
Replication
All phylogenetic trees were built using RAxML-NG v1.0.2 (46) assuming a general time-reversible model and a discrete gamma model of rate heterogeneity with 100 randomized parsimony starting trees and 1,000 bootstrap replicates (for details, see Data Accessibility).

Data Information	
Identifiers: Note that the raw sequencing reads of castanotis males A (SR00100) , B (Spain_1) , and F (Spain_2) were taken from ref . 4 (Sequence Read Archive accession numbers SRX6431677 to SRX6431681 , SRX6431686 , and SRX6431688 to SRX6431693) .	SRX6431677
	SRX6431681
	SRX6431686
	SRX6431688
	SRX6431693
Identifiers: The mtDNA tree was rooted by two guttata mtDNA assemblies SRA accession numbers SRR2299402 (47) and SRR3208120 (48), whereas the GRC haplotype tree was rooted at the midpoint of the two most distantly related haplotypes (Fig. 4A for haplotype network analysis, see SI Appendix, Results and Fig. S8).	SRR2299402
	SRR3208120

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Recombinant DNA			
FISH preparations were analyzed along the z-axis using a DSD2 confocal unit fitted to an Eclipse Ti inverted microscope (Nikon) with a Zyla sCMOS camera (Andor) under near-ultraviolet (405 nm) and green (550 nm) sequential excitation , obtained with a pE-4000 (CooLED) device , to determine whether the GRC signal from the GRC-carrying sperm was inside the nucleus .	pE-4000		
Software and Algorithms			
Images were processed using Fiji (33)	Fiji		Suggestion: Fiji (RRID:SCR_002285)(link)
To study the genetic diversity and the coevolutionary history of the GRC and its associated mtDNA , we sequenced (PCR-free) DNA from the testes and liver samples of a single male from each of the four major mtDNA haplotypes in captivity (i.e. , males C , D , and G to H see Samples) .	GRC		
Raw reads from all whole-genome sequencing libraries (PCR-free and Chromium) were processed following a modified version of the Genome Analysis Toolkit Best Practices Pipeline (34) .	Genome Analysis Toolkit		
We then mapped each library (paired-end reads) against the reference somatic genome , taeGut1 (36) , using BWA-MEM v0.7.17 (37) with the default settings while marking shorter split hits as secondary .	BWA-MEM		Suggestion: BWA-MEM2 (RRID:SCR_022192)(link)
In a next step , we used the Picard (38) MarkDuplicates option to mark mapped reads that might result from PCR duplication to reduce PCR bias in the abundance of certain DNA fragments during sequencing .	Picard		Suggestion: Picard (RRID:SCR_006525)(link)
For each library, we calculated read coverage using SAMtools v1.6 (39) depth per bp and used average values for each 1-kb window.	SAMtools		Suggestion: SAMTOOLS, RRID:SCR_002105(link)
Gene names were based on BLAT hits with at least one of the annotated	BLAT		Suggestion: BLAT (RRID:SCR_011919)(link)

genomes (zebra finch, chicken Gallus gallus, or human Homo sapiens).			
Finally, we used DnaSP v6.12.01 (45) to calculate the mean number of pairwise differences per site (p) for each GRC haplotype.	DnaSP		

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq